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22PH102 – ELECTROMAGNETISM AND MODERN PHYSICS

Unit-IV: MODERN PHYSICS_LM3

Consequences of the Special Theory of Relativity

❖ Length Contraction

Length contraction is a fundamental concept in Einstein's Special Theory of Relativity. It states that the length of an object appears shorter when it is moving relative to an observer. In this lecture material, we'll explore the theory behind length contraction, discuss its implications, and provide various examples to illustrate this phenomenon.

Length Measurement

- How do you measure the length of something?
  - If at rest, it is easy—just use a ruler ("proper length")
  - If moving with velocity v, a harder problem
  - Here is one way to do it.

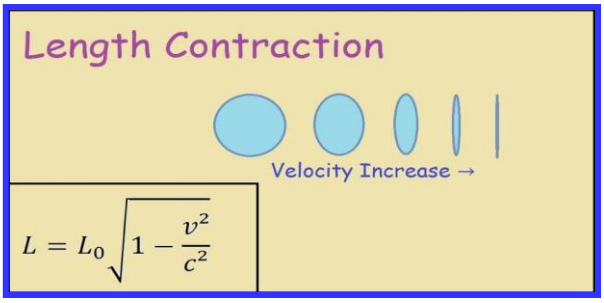
Length Contraction

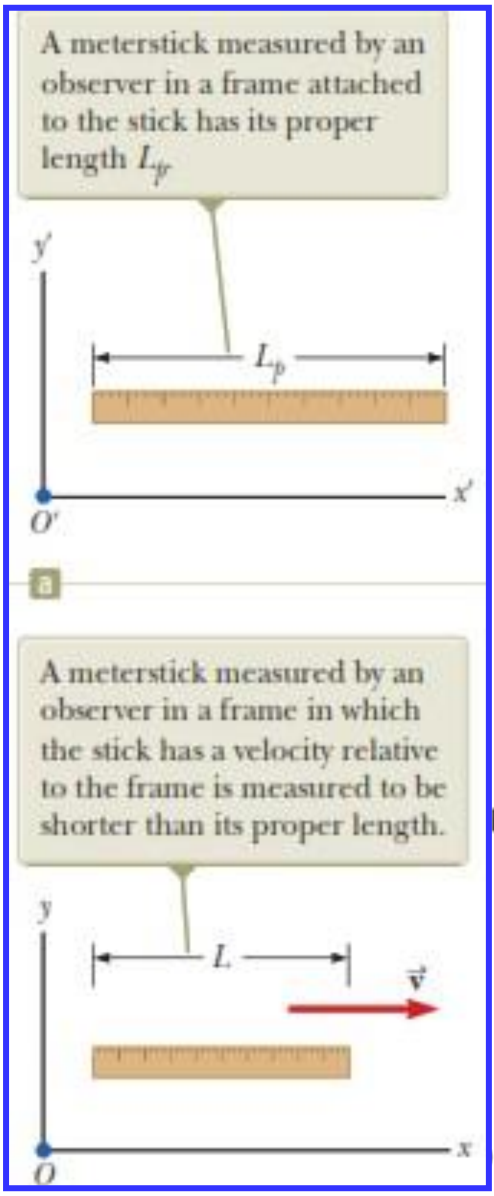
The measured distance between two points in space also depends on the frame of reference of the observer.

Proper Length = Length measured by an observer at rest relative to the object

The length of an object measured by someone in a reference frame that is moving with respect to the object is always less than the proper length

$$L = \frac{L_p}{\gamma} = L_p\sqrt{1 - \frac{v^2}{c^2}}$$





Length Contraction takes place only along the direction of motion

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train is **shorter in direction in motion** but just as high and wide as it was at rest

Example

- A rocket is travelling at 0·50 c relative to a space station. Astronauts on the rocket measure the length of the space station to be 0.80 km.

  Calculate the length of the space station according to a technician on the space station.

$$l' = l\sqrt{1-\frac{v^2}{c^2}} \qquad \therefore 0.8 = l\sqrt{1-\frac{0.5^2c^2}{c^2}}$$

Example 2

- A spaceship has a length of 220 m when measured at rest on the Earth. The spaceship moves away from the Earth at a constant speed and an observer, on the Earth, now measures its length to be 150 m.

  Calculate the speed of the spaceship in ms^{-1}.

$$l' = l\sqrt{1-\frac{v^2}{c^2}} \qquad 150 = 220\sqrt{1-\frac{v^2}{c^2}}$$

Space Travel

Alpha Centauri is 4.3 light-years from earth. (It takes light 4.3 years to travel from earth to Alpha Centauri). How long would people on earth think it takes for a spaceship traveling v=0.95c to reach A.C.?

$$\Delta t = \frac{d}{v} = \frac{4.3\text{ light - years}}{0.95\,c} = 4.5\text{ years}$$

How long do people on the ship think it takes?

People on ship have 'proper' time they see earth leave, and Alpha Centauri arrive. Δt_0

$$\Delta t = \frac{\Delta t_0}{\sqrt{1-\frac{v^2}{c^2}}} \qquad \Delta t_0 = \Delta t\sqrt{1-\frac{v^2}{c^2}} = 4.5\sqrt{1-.95^2}$$

$\Delta t_0 = 1.4$ years

Length Contraction

1. People on ship and on earth agree on relative velocity v = 0.95 c. But they disagree on the time (4.5 vs 1.4 years). What about the distance between the planets?

Earth/Alpha $L_0 = v\,t = .95\,(3\text{x}10^8\text{ m/s})\,(4.5\text{ years})$
$= 4\text{x}10^{16}\text{m}$ (4.3 light years)

Ship $L = v\,t = .95\,(3\text{x}10^8\text{ m/s})\,(1.4\text{ years})$
$= 1.25\text{x}10^{16}\text{m}$ (1.3 light years)

Length in moving frame

Length in object's rest frame

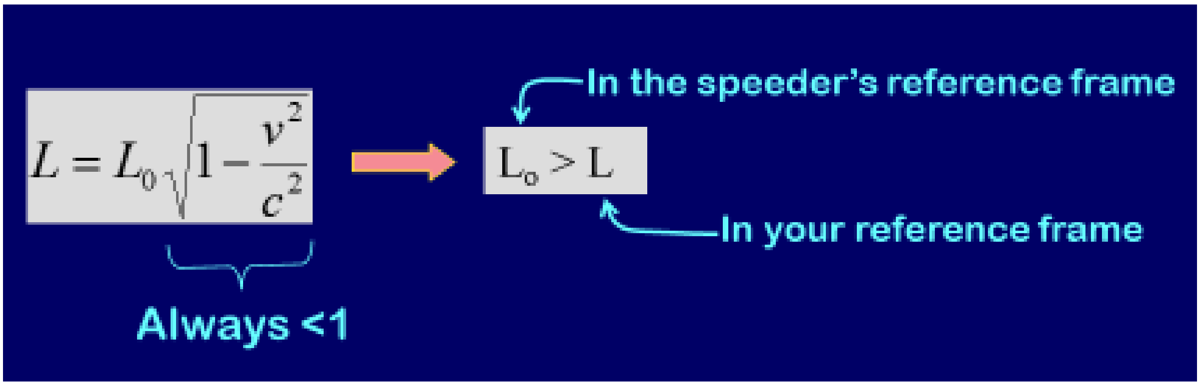
$$L = L_0\sqrt{1-\frac{v^2}{c^2}}$$

2. You're eating a burger at the interstellar café in outer space - your spaceship is parked outside. A speeder zooms by in an identical ship at half the speed of light. From your perspective, their ship looks:

(1) Longer than your ship

(2) Shorter than your ship

(3) Exactly the same as your ship



Comparison: *Time Dilation* vs. *Length Contraction*

- Δt_o = time in reference frame in which two events occur at same place "**proper time**"
  - i.e. if event is clock ticking, then Δt_o is in the reference frame of the clock (even if the clock is in a moving spaceship).

$$\Delta t_0 = \Delta t\sqrt{1-\frac{v^2}{c^2}} \rightarrow \Delta t > \Delta t_o$$

Time seems longer from "outside"

- L_o = length in rest reference frame as object "**proper length**"
  - length of the object when you don't think it's moving.

$$L = L_0\sqrt{1-\frac{v^2}{c^2}} \rightarrow L_o > L$$

Length seems shorter from "outside"

More About Proper Length

- Very important to correctly identify the observer who measures proper length.
- The proper length is always the length measured by the observer at rest with respect to the points.
- Often the proper time interval and the proper length are *not*

Implications of Length Contraction:

Length contraction has several profound implications, including:

Relativity of Measurement: Different observers in relative motion will measure different lengths for the same object.

FitzGerald-Lorentz Contraction: Length contraction was initially proposed by George FitzGerald and independently by Hendrik Lorentz to explain the null result of the Michelson-Morley experiment, which aimed to detect the Earth's motion through the luminiferous ether.

Examples of Length Contraction:

Example 1: Moving Rod

Consider a rod moving at a significant fraction of the speed of light relative to an observer. From the observer's perspective, the length of the rod contracts in the direction of motion. This contraction is observable and measurable.

Example 2: Spacecraft Design

In designing spacecraft that travel at relativistic speeds, engineers must account for length contraction effects. For instance, the distance between components of the spacecraft may appear shorter to observers on Earth than to the astronauts aboard the spacecraft. Engineers must compensate for this contraction to ensure the proper functioning and safety of the spacecraft.

Example 3: Particle Colliders

Particle accelerators, such as the Large Hadron Collider (LHC), accelerate particles to near the speed of light. As particles approach these speeds, they experience length contraction. This contraction effectively reduces the distance traveled by particles within the accelerator, allowing them to collide with higher energies and enabling scientists to study fundamental particles and forces.

Example 4: Cosmic Ray Showers

Cosmic ray showers, generated by high-energy particles from outer space, produce secondary particles that travel at relativistic speeds. As these particles travel through Earth's atmosphere, they experience length contraction. This contraction affects the observed dimensions and trajectories of the secondary particles, influencing the characteristics of cosmic ray showers observed by detectors on the ground.